Marc Pinet

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available upon request https://marcpinet.fr/



PhD Researcher with 2 years of experience in industry R&D and academic research. Specialized in Machine Learning, Anomaly Detection, Time Series Analysis, and Large Language Models (LLMs). Contributor to multiple open-source projects, with international experience across France, Canada, South Korea and Vietnam.

Professional Experience

2025 – 2028 **PhD Researcher in Machine Learning**, Orange Research & LIG & CNRS.
Detection and explanation of anomalies in time series data for network security. Anomaly detectors flag that *something* happened, but never *what*. My thesis addresses this gap, from detecting anomalies to characterizing and explaining them in natural language, sparing experts the manual investigation of each alarm and reducing intervention costs.

2024 – 2025 **Machine Learning Engineer Apprentice**, SAP Labs France.
Unsupervised anomaly detection system for heterogeneous data (logs, metrics, traces) using pattern variation detection, and time series correlation. Optimized real-time pipeline reducing memory by 70% while handling 10M+ events/day. Built RAG-enhanced LLM agent for automated anomaly reports with root cause analysis.

2024 – 2024 **Data Scientist Intern**, SAP Labs France.
Time series trends analysis using tools such as Prophet, SARIMA, STL, and DTW. Identified correlations between logs and metrics to predict system behaviors. Built a robust pipeline for live data retrieval from internal APIs with integrated data preprocessing.

2022 – 2022 **IoT Developer Intern**, Da Nang International Institute of Technology, Vietnam.
Developed a LoRaWAN-based emergency messaging system, covering 25km range, ensuring emergency communication in remote areas without internet access.

Education

2025 – 2028 **PhD in Computer Science & Mathematics**, Université Grenoble Alpes.
Thesis: *Self-supervised deep learning detection and explanation of anomalies in time series.*

2022 – 2025 **Engineer's Degree in Computer Science**, Polytech Nice Sophia.
Specialty: *Artificial Intelligence & Data Engineering*.

2022 – 2022 **Exchange Semester**, Université du Québec à Chicoutimi (UQAC), Canada.
In partnership with IUT Nice Côte d'Azur to reward my results and complete my diploma.

2020 – 2022 **Associate of Science in Computer Science**, IUT Nice Côte d'Azur.

Skills

Programming	Python, Java, C/C++, SQL, Bash, Research & Software Engineering
Artificial Intelligence (AI)	Machine Learning, Deep Learning, Anomaly Detection, XAI, RAG, LLMs, Time Series Analysis, Data Science, Self-Supervised Learning, Representation Learning, Knowledge Graphs. **Frameworks:** PyTorch, TensorFlow, JAX, Hugging Face Transformers, Scikit-learn.
MLOps & Cloud	Weights & Biases, MLflow, FastAPI, Docker, AWS, GCP, Azure, Git.
Other Skills	Scripting, Databases, Communication, Technical Writing, Teamwork.
Languages	French (Native), English (C1), Spanish (B1), Chinese (A1).

Publications

1. M. Pinet, J. Cumin, S. Berlemont, and D. Vaufreydaz, "Anomalies in Multivariate Time Series Benchmarks Are Mostly Univariate," in *KDD 2026, Mining and Learning from Time Series Workshop (MILETS)*, **rank: A***, Jeju, South Korea, presented orally, Jun. 1, 2026. DOI: 10.48550/arXiv.2606.02670

Service

Sep–Oct 2026 **Reviewer**, FMTS @ NeurIPS 2026.

Jun–Dec 2026 **Ethics Reviewer**, NeurIPS 2026.

Relevant Open-Source Projects

1. M. Pinet, *therightbibtex*, Finds the right BibTeX entry for any paper by querying 15+ bibliographic sources in parallel (DBLP, OpenReview, Crossref, arXiv, Semantic Scholar...), grouping the versions of the same paper and recommending the published one with per-field provenance. CLI and local web interface, published on PyPI, Oct. 2026. URL: https://github.com/marcpinet/therightbibtex
2. M. Pinet, *ZoteGrep*, Full-text search engine for Zotero libraries: phrase, proximity (NEAR) and regex queries over every PDF, note and annotation, answered with the exact sentence and page number. SQLite FTS5 trigram index, OCR of scanned pages and relevance ranking, as a CLI and a local web interface, published on PyPI, Oct. 2026. URL: https://github.com/marcpinet/zotegrep
3. M. Pinet, *jobgpumonitor*, Zero-dependency monitoring of GPU jobs on HPC clusters (Slurm, OAR, PBS, LSF): the job streams its progress, GPU utilization, memory and tracebacks as JSONL events, and a companion server sends alerts (crash, OOM, time-out, idle GPU) to ntfy, Telegram or webhooks. Published on PyPI, Sep. 2026. URL: https://github.com/marcpinet/jobgpumonitor
4. M. Pinet, *Citracer*, Trace citation chains for any keyword across research papers and render them as an interactive graph, Apr. 2026. URL: https://github.com/marcpinet/citracer
5. M. Pinet, *MIMIMI*, Self-supervised complex neural network for machine sound anomaly detection using phase-aware spectral analysis. PyTorch implementation of EUSIPCO 2021 paper achieving 95.18% AUC on MIMII dataset, Jul. 2025. URL: https://github.com/marcpinet/mimimi
6. M. Pinet and A. Rodriguez, *Olympics Knowledge Graph*, A semantic web project enriching Olympics data through spaCy's deep learning models and knowledge graphs. Features enrichments through APIs, SPARQL inference rules, and SHACL constraints. Expanded from 355 to 15,000+ triples with DBpedia/Wikidata linking, Jan. 2025. URL: https://github.com/marcpinet/websem-og24
7. M. Pinet, *Neuralnetlib*, A flexible machine & deep learning framework built from scratch using only NumPy. Supports a large amount of network architectures, layers, activations, losses, optimizers and ML tools, Nov. 2024. URL: https://github.com/marcpinet/neuralnetlib
8. M. Pinet, *Handigits*, Background-independent deep learning model for hand sign digit recognition. Used my own framework (Neuralnetlib) for model training and inference, and Google MediaPipe for hand tracking and preprocessing, May 2024. URL: https://github.com/marcpinet/handigits
9. M. Pinet, *EdgeAI: Bird Species Detection*, CNN-based bird species audio recognition, optimized using TinyML on STM32 microcontroller with real-time I2S audio processing, 16-bit fixed-point quantization, and LoRaWAN IoT connectivity achieving 86% accuracy with optimized memory footprint for embedded deployment, Apr. 2024. URL: https://github.com/marcpinet/edgeai-bird
10. M. Pinet, *Neat Cars*, Autonomous vehicle AI agents leveraging NEAT genetic programming and evolutionary reinforcement learning to optimize neural network topologies with real-time visualization and multi-generational evolution, Mar. 2023. URL: https://github.com/marcpinet/neat-cars

11 M. Pinet, *Epidemic Modeling*, Visual simulation of a pandemic spread, following the SEIRD model, with statistical summary graphs based on simulated data. Validated against research literature with real-time intervention scenario analysis, Dec. 2021. URL: https://github.com/marcpinet/epidemic-modeling

Miscellaneous Experience

Certifications

2026 **Research Integrity and Ethics (University of Bordeaux)**.

2025 **Stanford | DeepLearning.AI Deep Learning Specialization (Coursera)**.

Stanford | DeepLearning.AI Machine Learning Specialization (Coursera).

2024 **TOEIC** – Score: 950/990.

2023 **Quantum Computing and Information Summer School (EIT Digital)**.

2021 **Certification Voltaire** - Spelling: 975/1000, Writing: 8/9.

Licenses

2024 Boating License.

2022 Driver's License.